

COM410 Programming in Practice

Case Study

Testing Summary

Three testing techniques have been employed as described below.

1. JUnit Testing

The **VehicleTest** class uses a set of JUnit tests to check that the correct score is assigned to a selection of vehicle objects. A version of the constructor was implemented that allows a vehicle type and colour to be assigned rather than randomly generated, and this was used to confirm that non-red vehicles were assigned the appropriate score according to the specification.

2. Test Code

The **VehicleTest** class also contains a test method to measure the distributions of vehicle types and colours so that they can be compared against the distributions in the specification. 1,000,000 random vehicles are generated and a HashMap is used to keep a count of the number of times each type and colour are generated. The results are presented as a percentage of the total vehicles generated.

3. Observation Testing

Remaining functionality was tested by running the application and observing the results. Specifically, the following elements of functionality were tested.

Functionality	Expected Outcome	Result
Player names	User is prompted to enter 6 player names	Pass
Player sequence	Team 1 plays before team 2 and within a team, each player's turn is in order	Pass
Gameplay	A player's turn is ended by the first red vehicle of any type that they observe	Pass
Player scoring	A player's score is the sum of the vehicle scores that were observed	Pass
Team scoring	A team's score is the sum of the individual players' scores	Pass
Winner	The team with the highest cumulative score is declared as the winner (or the match is drawn if team scores are equal)	Pass